WHEN DO AGENT HARNESS IMPROVEMENTS GENERALIZE? SEMANTIC BINDING, EXTERNAL VALIDITY, AND EVALUATION RELIABILITY IN LLM AGENTS

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ABSTRACT

A harness intervention can survive controlled development and a pre-frozen within-family held-out evaluation without establishing transfer across models or semantic task families. On a controlled tool-grounded *semantic-handoff* task, a non-answer-bearing "clarity bundle" (BundleS) suppresses a specific tool-green semantic-binding failure for Qwen3.7-Max, confirmed on a pre-frozen held-out instance (Study I). But the effect does not transfer: under exact counterbalancing it is not established for DeepSeek-V4-Pro, and on two structurally independent task families it is a clean null or a non-discriminative ceiling (Study II). When we expand an external STA family prospectively from three to twelve independently frozen instances, the descriptive direction of the comparison reverses while still failing to establish a stable benefit. We contribute a *Harness Effect Audit Protocol*—four layers (capability, sampling, execution, artifact integrity) instantiated and stress-tested in this study—each shown catching a real threat; an independent benchmark-repair audit (Terminal-Bench 2.0→2.1) shows broad but predominantly partial correspondence (22 of 26 repairs had at least partial correspondence, only one directly captured). Our controlled task instances test existence, recurrence, and transfer direction rather than estimate population-level pass rates. A preregistered blinded human construct-validity study is reported as unexecuted. The contribution is an evaluation study, not a proposed improved harness.

1 INTRODUCTION

A harness intervention can survive controlled development and a pre-frozen within-family held-out evaluation without establishing transfer across models or semantic task families. This is the result that motivates the paper: an effect that looks robust under repeated within-condition replication and a frozen held-out instance can still fail the first strict external-validity test—and, as we show, can even reverse direction when an external family is expanded prospectively.

We study this on tool-grounded EDA workflows, where an LLM agent binds quantities to canonical typed roles and submits an artifact to a real commercial tool. On a *semantic-handoff* task the agent assigns a tuple to canonical typed roles from heterogeneous, role-misleading evidence; a wrong binding is not a syntax error—it produces a *plausible green signoff*—so only a typed provenance/authority grader distinguishes correct from wrong. We ask: **RQ1** can harness information structure change semantic-binding behavior (controlled setting)? **RQ2** does it transfer across models and task families? **RQ3** which evaluation controls make these conclusions valid?

Contributions. (1) A controlled semantic-binding mechanism + typed failure taxonomy, isolated from analog-design difficulty. (2) The transfer-failure-after-held-out-confirmation finding, including a prospective 12-instance STA expansion whose descriptive direction reversed relative to the three-instance pilot while still establishing no treatment effect. (3) The Harness Effect Audit Protocol, instantiated and stress-tested here, plus an independent benchmark-repair audit as external taxonomy application.

Novelty (disciplined). Prior work establishes that harness configurations affect agent performance (Yao et al., 2026; Wang et al., 2026; Ursekar et al., 2026). We instead study *what evidence is required to attribute such an effect, whether within-family held-out confirmation predicts transfer, and how execution-layer validity failures can change the resulting scientific conclusion.* We do not propose an improved harness, and we do not claim harness improvements generally fail to transfer—indeed Ma et al. (2026) report cross-benchmark harness gains, which strengthens our claim that transfer is an empirical property to be measured rather than assumed.

2 RELATED WORK AND PROBLEM FORMULATION

Harnesses and harness effects. A *harness* is the task-level information structure visible to the agent (prompt, visible files, disclosure bundle, public tool feedback, action surface). A *harness effect* is a change in semantic-binding correctness attributable to harness information content, holding the hidden truth, grader, and tool fixed.

Problem formulation. Each task is a *semantic handoff*: bind a tuple to canonical typed roles from role-misleading evidence. The workflow family binds PVT descriptors to typed axes; Family A (STA/PrimeTime) binds (intent_class, target_partition, check_mode) from an authority provenance DAG; Family B (SPICE/HSPICE) binds (corner, load_condition, metric) from a request–authority join. Families A/B are structurally independent from the workflow family (verified under five pre-registered criteria; operational, not a proof).

Positioning. Yao et al. (2026) broadly characterize model×harness interactions; Wang et al. (2026) critique harness-leaderboard conflation; Ursekar et al. (2026) optimize harnesses; Zhu et al. (2025) and Shao et al. (2026) address benchmark setup, shortcut/reward validity, and protocol validity. We differ by isolating a controlled semantic-binding mechanism, testing transfer after held-out confirmation, and auditing transport/membership/integrity controls during executable evaluation (Table 1). Two further distinctions: **TAB** (Mavali et al., 2026) studies the selective use of necessary environmental cues in the presence of plausible distractors, whereas our construct is role-conditioned semantic binding and provenance/authority attribution—including cases where evidence is available but values are bound to the wrong role while the tool stays green. Ma et al. (2026) report cross-benchmark harness gains, which we use as evidence that some interventions do transfer, so transfer must be demonstrated rather than inferred.

Per-work gap. Each prior line leaves a specific opening. Yao et al. (2026) measures harness effects at scale but does not isolate a single semantic mechanism or test transfer after held-out confirmation. Wang et al. (2026) identifies overfitting and leaderboard conflation but does not construct independent task families to test whether a within-family result generalizes. Ursekar et al. (2026) treats the harness as an optimization target, not as a scientific claim whose attribution must be audited. Zhu et al. (2025) and Shao et al. (2026) sharpen benchmark-construction and protocol-validity standards but do not report a per-control threat log showing, for executable agent evaluation, which measurement failures would have changed the headline. Our contribution sits in those gaps: a controlled mechanism whose failure stays tool-green, an external-validity test that held-out confirmation did not survive, and an audit protocol whose controls are shown to be causal.

3 HARNESS EFFECT AUDIT METHODOLOGY

Tasks and tool grounding. All families run on real commercial tools (PrimeTime; HSPICE) through a transparent remote shim—not tool simulations. A wrong binding produces a plausible green signoff/number, so tool success is necessary but not sufficient; only the typed provenance/authority grader rejects a wrong binding.

Conditions. **Base** (ambiguous handoff; no answer-bearing disclosure); **BundleS** (canonical labels C1, value-domain definitions C2, glossary C4, procedural contract C7; no answer-bearing C6; no golden values); **TypedContract** (same information as BundleS as a JSON Schema). Hidden truth and grader are identical across conditions.

Table 1: Positioning relative to five lines of related work.

Prior work	What it establishes	What this paper additionally does
Harness-Bench (Yao et al., 2026)	Broad model×harness characterization	Controlled semantic-mechanism isolation + typed taxonomy; claim-validity analysis
Rethinking Harness Eval. (Wang et al., 2026)	Leaderboard conflation; search/overfitting	Held-out confirmation still need not transfer cross-family
HarnessOpt-Bench (Ursekar et al., 2026)	Optimizer capability	Studies validity of harness-effect claims, not optimization
ABC (Zhu et al., 2025) / Best-Practices	Setup; shortcut/reward validity	Additionally audits transport, membership, tool health, action surfaces, canonical integrity
Protocol validity (Shao et al., 2026)	Protocol design; reward-hacking surface	Measurement validity as the contribution; per-control threat log

Construct: what "semantic correctness" means (and does not). Semantic correctness is *not* inferred from tool success, from a hidden numeric answer, or from the artifact score. It is decided solely by whether the submitted tuple matches the binding attested by the independent provenance/authority oracle. *Worked example (Family A):* on one instance the visible evidence attests `intent.md` → intent_class=*functional_close*, the CDC report → target_partition=*core*, and the coverage matrix → check_mode=*setup*. The agent submitted (functional_close, core, *both*): PrimeTime returns a green signoff, the value is type-valid, and the artifact is partially scored—yet it is *semantically wrong*, because the coverage authority attests *setup*, not *both*; the oracle rejects it as a role-conditioned value-selection failure. This construct is instantiated identically (different typed roles and authority sources) in the workflow family (PVT axes), Family A (intent/partition/check_mode), and Family B (corner/load/metric), each with its own independent grader implementation. We do *not* claim human validation of these labels (Study B was preregistered but unexecuted; see Limitations).

Design. Seeded exact-counterbalanced blocked randomization, frozen before any paid call. Task instance is the primary unit; stochastic repetitions are nested. We report paired directions and raw counts per instance and do not pool trajectories into a single significance headline.

Detectability. With small instance counts our design is diagnostic rather than population-estimating: it tests existence, recurrence, ceiling/null behavior, and transfer direction; it cannot resolve small effect sizes. We do not use small-n bootstrap significance or a trajectory-pooled p-value to compensate.

Models and scope. We evaluate Qwen3.7-Max (a frontier reasoning model that was responsive to the clarity bundle in pilot pairing) and DeepSeek-V4-Pro (a second frontier model for the cross-model test). These were chosen for availability through our gateway and for contrasting reasoning profiles; they are *not* presented as representative of the frontier-model population, and the transfer conclusions are scoped to the models tested.

Why the design isolates the harness variable. The controlled pair shares identical hidden truth, grader, and tool; it differs only in the disclosure bundle. Because the design is marginally timed, every binding—correct or wrong—returns a green signoff, so the tool cannot be the operative variable. The typed oracle, not the tool, decides correctness, and the oracle is held fixed across conditions. The only thing that changes between Base and BundleS is what the agent can read. This is what lets us attribute a behavioral change to the harness information structure rather than to task difficulty, tool behavior, or grader leniency. The failure taxonomy then makes the changed behavior *visible*: a tool-green signoff that the oracle rejects as an axis-binding or value-selection failure is exactly the kind of outcome an aggregate pass@k metric would count as success, dissolving the effect.

Audit protocol (preview). Four layers, each reported independently: (1) capability validity, (2) sampling validity, (3) execution validity, (4) artifact integrity. The protocol is instantiated and stress-tested in this study—we do not claim it is broadly validated or universally reusable.

Experimental status. Table 2 records the registration type of each result; only predeclared, frozen-held-out, exact-counterbalanced, and prospective-confirmatory results are headline.

Table 2: Experimental-status registration.

Registration type	Result (used in)	Headline?
Predeclared (controlled pair)	Workflow: clarity bundle suppresses axis binding (Study I)	Yes
Pre-frozen held-out	Workflow: BundleS on a pre-frozen held-out truth (Study I)	Yes
Exact-counterbalanced	Cross-model: BundleS vs Base, balanced (Study II)	Yes
Prospective confirmatory	STA external family: 12 pre-frozen new instances (Study II)	Yes
Secondary extension (pre-frozen)	TypedContract; SPICE ceiling (Study II)	Yes (scoped)
Exploratory (sequential)	Component localization C1/C2/C4/C7/C24 (Appendix)	No
Preregistered but unexecuted	Blinded human construct-validity study (Study B)	n/a

4 Study I — Controlled Semantic-Binding Effects (RQ1)

Question: can harness information structure change semantic-binding behavior, holding truth and grader fixed? On the workflow controlled pair (identical truth/grader; ambiguous vs. clear handoff), the full clarity bundle suppresses the axis-binding failure for Qwen3.7-Max: Base 1/3 (2 axis-binding) $\to$ BundleS 3/3 (0 axis-binding). The schema/contract subset BundleS (excluding the answer-bearing C6) suffices on the development pair, and generalizes to a *pre-frozen held-out* instance: BundleS 3/3 vs Base 1/3. The failure the bundle suppresses (Figure 1) partitions the workflow ledger (54 primary episodes) into 29 correct, 20 *axis-binding*, and 5 *role-conditioned value-selection* failures—all tool-green, rejected only by the typed oracle. **RQ1 is supported within the controlled workflow family for Qwen3.7-Max; we make no universal improvement claim.**

No stable minimal component (exploratory). A sequential decomposition tested whether a stable minimal subset underlies BundleS (C1+C2+C4+C7). None was isolated: C1 alone did not eliminate axis errors; C2-only and C4-only did not reproduce the bundle pattern; a predeclared in-window C24 bridge failed its $\geq$3/4-with-0-axis threshold (2/4, 1 axis). The bundle-level effect is real within-family for Qwen3.7-Max, but the operative sub-component is not identified—the C2$\times$C4 joint-effect hypothesis remains unresolved rather than refuted. This bounds the mechanistic claim: we know the *non-answer-bearing* disclosure is operative (not C6), but not which minimal combination suffices.

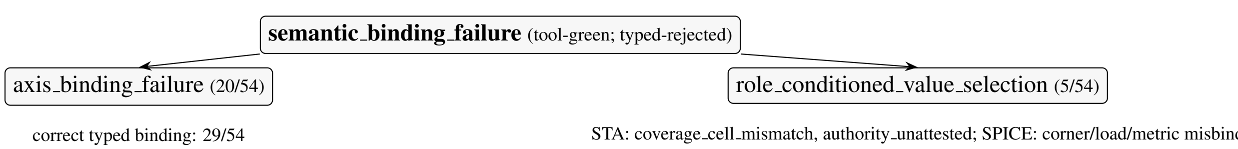

Figure 1: Semantic-binding failure taxonomy (all failures tool-green, oracle-rejected).

5 Study II — Model and Task-Family Transfer (RQ2)

Question: does the within-family effect transfer across models and families? Table 3 is the main result. The RQ2 headline separates three outcomes: *null/unestablished transfer* (STA, discriminating; DeepSeek cross-model), *ceiling* (SPICE, non-discriminative), and *direction reversal under prospective expansion* (STA pilot vs. prospective).

STA—prospective confirmatory batch (authoritative). Across 12 pre-registered new instances, BundleS vs Base produced 3 improving, 2 declining, and 7 tied instances; the exact sign-test $p = 1.0$ and the pre-registered instance-level permutation sensitivity $p = 0.31$. *The three-instance pilot suggested Base $\geq$ BundleS, whereas the prospectively frozen twelve-instance batch reversed the descriptive ordering but still did not establish a treatment effect. We therefore treat the reversal as*

Table 3: Main result matrix (semantic-binding correct rate; instance is the unit).

Family / model	Base	BundleS	TypedContract	n	verdict
Workflow / Qwen3.7-Max (development)	1/3	3/3	—	1	controlled effect
Workflow / Qwen3.7-Max (frozen held-out)	1/3	3/3	—	1	held-out confirmed
Workflow / DeepSeek-V4-Pro (counterbalanced)	3/4	3/4	—	1	no benefit
STA / Qwen3.7-Max (prospective)	**0.208**	**0.333**	**0.458**	**12**	**transfer not established**
SPICE / Qwen3.7-Max	1.00	1.00	1.00	3	ceiling (non-discriminative)

evidence against trusting direction from very small task panels, not as evidence that BundleS transfers. We do not claim a significant BundleS improvement, a significant TypedContract improvement, or a stable STA negative. TypedContract is descriptively highest (0.458) but *no preregistered TypedContract advantage was established.* The historical three-instance pilot (Base 0.50 / BundleS 0.33 / TypedContract 0.33) is reported separately as descriptive pilot evidence and is **not pooled** with the prospective $n = 12$.

Per-instance prospective evidence (Table 4). The prospective task is hard: seven of twelve instances sit at the floor (0/0/0) across all three conditions, so the family-level statistics effectively hinge on the five discriminating instances (0010–0012, 0014–0015). The condition differences are concentrated there—e.g., instance 0011 moves Base 0/2 $\rightarrow$ BundleS 2/2 $\rightarrow$ TypedContract 2/2, while 0004 moves Base 1/2 $\rightarrow$ BundleS 0/2. Within-instance two-repetition agreement is high (only one Base cell, instance 0004, disagrees with itself across repetitions), so between-instance spread—not run-to-run noise—drives the family means. This is stability of direction at fixed instance, not a population estimate.

Table 4: Prospective STA per-instance outcomes (n=12; 2 nested reps; Y=correct, n=wrong).

	2-rep outcome (rate)			paired diff		
instance	Base	BundleS	TypedContract	S−Base	TC−Base	TC−S
p15_eval_0004	nY (.5)	nn (.0)	nn (.0)	−.5	−.5	.0
p15_eval_0005	nn (.0)	nn (.0)	nn (.0)	.0	.0	.0
p15_eval_0006	nn (.0)	nn (.0)	nn (.0)	.0	.0	.0
p15_eval_0007	Yn (.5)	nn (.0)	nn (.0)	−.5	−.5	.0
p15_eval_0008	nn (.0)	nn (.0)	nn (.0)	.0	.0	.0
p15_eval_0009	nn (.0)	nn (.0)	nn (.0)	.0	.0	.0
p15_eval_0010	nY (.5)	YY (1)	YY (1)	.5	.5	.0
p15_eval_0011	nn (.0)	YY (1)	YY (1)	1.0	1.0	.0
p15_eval_0012	nn (.0)	YY (1)	YY (1)	1.0	1.0	.0
p15_eval_0013	YY (1)	YY (1)	YY (1)	.0	.0	.0
p15_eval_0014	nn (.0)	nn (.0)	Yn (.5)	.0	.5	.5
p15_eval_0015	nn (.0)	nn (.0)	YY (1)	.0	1.0	1.0

Why the direction reversal matters. The pilot (Base 0.50 $\geq$ BundleS 0.33) and the prospective batch (Base 0.208, BundleS 0.333) disagree on the *sign* of the comparison; neither is statistically distinguishable from no effect. The reversal is therefore not evidence of a mechanism change—it is a direct demonstration that three task instances cannot pin even the direction of a harness comparison on this family. Per-instance, the prospective task is hard: many instances sit at the floor across all three conditions, and condition differences concentrate in a handful of discriminating instances; within-instance two-repetition agreement is high (only one of twelve Base cells disagrees with itself across repetitions), so between-instance spread, not run-to-run noise, drives the family means.

Cross-model + SPICE. Under exact counterbalancing the within-family BundleS benefit is not established for DeepSeek-V4-Pro (3/4 = 3/4). Family B (SPICE) is at ceiling (1.00 across conditions)—non-discriminative, not a refutation.

Three outcomes, kept distinct. The transfer evidence falls into three categories that must not be conflated. (i) *Null on a discriminating family* (STA): the measurement leaves room to improve, so the absence of an effect is informative—but at $n = 12$ it cannot rule out a small effect, and its sign disagrees with the pilot. (ii) *Ceiling* (SPICE): Base already solves the binding, so the treatment has no headroom; this is a measurement limitation, not evidence against transfer. (iii) *Direction reversal under prospective expansion* (STA pilot vs. prospective): an illustration that very small panels cannot fix even the sign of a comparison, not a mechanism finding. Conflating (i) and (ii) into a single "no transfer" claim would overstate the SPICE evidence; treating (iii) as a real effect would overstate the STA evidence. We therefore report each separately.

RQ2. *Within-family held-out confirmation did not establish cross-model or cross-family transfer.*

6 Study III — Evaluation Validity and the Independent Benchmark Repair Audit (RQ3)

Each headline conclusion could have been manufactured or masked by an evaluation artifact. The four-layer protocol is instantiated and stress-tested in this study; Table 5 lists concrete incidents where a control *materially changed the interpretation* (not infrastructure history).

Table 5: Audit-protocol incidents: each control changed what the scientific conclusion would otherwise have been.

Threat	Audit / control	Conclusion that would have been wrong or ambiguous
Non-streaming censoring (L3)	SSE streaming + terminal-transport arbiter	Qwen3.7-Max would have scored as failing the controlled pair (a false negative capability result)
Position confounding (L2)	Exact position-balanced blocking	Base/BundleS order could have leaked into the contrast
Recovered transport degradation (L3)	Terminal-valid vs recovered split; validity-only replace	A hard-but-recovered solve could have been discarded or mis-scored
PrimeTime corrupted measurement (L3)	Tool-health sentinel + full-path bookends	A tool-server hiccup could have read as a model failure
SPICE action-surface false positive (L4)	Immutable core / derived deck + forensic audit	All 12 SPICE episodes would have been zeroed (a false ceiling/refutation)
Canonical-source mutation (L4)	Exact-commit worktree + frozen-hash guard	A silently rewritten golden would have read as a $\sim$day-long tool outage

Independent Benchmark Repair Audit (Terminal-Bench 2.0→2.1). As external taxonomy application, we coded the 26 officially documented repairs under the four-layer-plus-Other rubric, *frozen before inspecting the repairs*. We define: **direct** = our protocol's layer would detect/prevent this repair's issue; **partial** = a protocol layer relates conceptually but would not necessarily catch this specific case; **not-covered (Other)** = outside the four layers. Result: **1 direct / 21 partial / 4 not-covered** of 26; layers capability 17, sampling 0, execution 10, artifact 1; 6 multi-layer. *The preregistered taxonomy showed broad but predominantly partial correspondence with independently documented benchmark repairs: 22/26 had at least partial correspondence, but only one was directly captured and four lay outside the taxonomy.* This is **static retrospective taxonomy application**, not runtime external validation, not evidence that the protocol predicted or would necessarily have prevented the repairs; and the absence of sampling-layer repairs (L2 = 0) means this audit provides no external support for that layer. The full 26-task coding + source evidence is in the Appendix.

What the coding means (examples). The single *direct* case is `fix-git`: Terminal-Bench removed install files that agents could access to cheat—precisely an action-surface/reward-hacking threat our L4 addresses. *Partial* cases are instructive: instruction/verifier misalignments (e.g., `polyglot-c-py`, where a test rejected legitimate compilation byproducts) map conceptually to capability validity (L1) but our L1 controls target tool-green semantic binding, not generic verifier over-specification; resource/timeout raises (e.g., `caffe-cifar-10`) map to execution validity (L3) but our L3 is PT/HSPICE tool-health, not container CPU caps. The four *not-covered* cases are package-pin removals—benchmark-environment maintenance outside our four layers. The L2 (sampling) count of zero is itself informative: the Terminal-Bench revision did not touch sampling/counterbalancing, so this audit neither supports nor refutes our sampling layer—we state this limitation rather than claim blanket validation.

Why the controls are causal, not cosmetic. Each incident in Table 5 changed the scientific conclusion, not merely the engineering record. Without SSE streaming, Qwen3.7-Max would have been recorded as failing the controlled pair—inverting RQ1 from "effect present" to "effect absent." Without the SPICE action-surface repair, all twelve SPICE episodes would have been zeroed, manufacturing a false ceiling that would have read as either a refutation or a trivial success depending on the reader. Without the canonical-hash guard, a silent golden rewrite would have masqueraded as a day-long tool outage, contaminating every tool-health judgment in that window. These are not implementation footnotes: each would have altered the headline interpretation, which is why the audit protocol is reported as a first-class methodological contribution rather than a lab notebook.

7 Limitations and Discussion

Controlled executable tasks, not sampled production incidents. EDA-heavy environments—the transfer result is established only within EDA. *Small instance counts*—the design is diagnostic, not population-estimating; the STA prospective null cannot rule out a small effect, and the pilot-to-prospective direction reversal illustrates small-sample instability rather than estimating a rate. *Construct validity rests on executable provenance/authority oracles rather than independent blinded human judgment*: the preregistered human construct-validity study (Study B) is **preregistered but unexecuted because qualified independent human annotators were unavailable**; no LLM annotator was substituted, and we do not claim independent human validation of the taxonomy. *Model scope*—Qwen3.7-Max and DeepSeek-V4-Pro are not representative of the frontier-model population; conclusions are scoped to them. *Audit protocol*—instantiated and stress-tested here, not broadly validated; the Terminal-Bench audit is retrospective taxonomy application, not prediction/prevention. *No mechanism isolated*—the bundle-level effect is real within-family for Qwen3.7-Max but no stable minimal component was identified (Appendix). These limitations bound but do not undermine the central claim: the paper does not assert that BundleS works, fails, or transfers as a population property—it asserts that the evidence required to make any of those assertions was not met by within-family held-out confirmation alone, and it shows the measurement controls needed to know the difference.

Discussion. The result is a caution and a method. The caution: a single within-family replication—even a frozen held-out one, and even a small external pilot—is weak evidence of generalization for harness effects. The method: the audit protocol makes the positive and null conclusions independently checkable. Some harness interventions do transfer across benchmarks (Ma et al., 2026); transfer is therefore an empirical property that must be demonstrated rather than inferred from within-family confirmation.

Broader implications. Beyond EDA, the four-layer audit protocol is a checklist that any agent-benchmark team can apply: does a wrong answer score tool-green (L1)? Are trajectories repeated and counterbalanced (L2)? Can a transport or tool failure masquerade as a capability result (L3)? Can an action-surface artifact manufacture or mask a score (L4)? Each question is concrete, and each has a documented incident in which ignoring it would have changed the paper's headline. The Terminal-Bench audit shows that independent benchmark teams confront overlapping (predominantly L1 and L3) validity threats, suggesting the taxonomy captures real concerns—but the limited direct correspondence (1/26) and the absence of L2 support honestly bound the protocol's current external validation. The framework's value is not that it solves benchmark validity, but that it makes the validity of each claim checkable, reproducible, and separable.

Reproducibility. Deterministic generators (fixed seeds); independent per-family graders; exact-commit isolated-worktree execution with pre/post-episode canonical-hash verification; committed sample-membership arbitration; validity-only replacement; per-episode custody byte-matching; sanitized custody evidence. Frozen manifests, the treatment mapping, schedules, and provenance are in anonymized supplementary material. Program totals: 58 (workflow) + 24 (cross-family core) + 36 (cross-family secondary) + 72 (prospective STA) paid primary episodes; committed-ledger cost ¥745.29.

8 Conclusion

A harness intervention can survive controlled development and a pre-frozen within-family held-out evaluation without establishing transfer across models or semantic task families; prospectively expanding an external family from three to twelve instances reversed the apparent comparison direction while still not establishing a stable benefit. We contributed the Harness Effect Audit Protocol (four layers, stress-tested) and applied it retrospectively to 26 independent benchmark repairs (broad but predominantly partial correspondence). The contribution is an evaluation study—of *when* an effect is attributable, *whether* it transfers, and *which* controls make either claim credible—not a proposal of an improved harness.

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A Study I — per-instance ledger and minimal-component ablation

Workflow ledger (54 primary episodes). 29 correct typed bindings, 20 *axis-binding* failures, 5 *role-conditioned value-selection* failures. Family-specific subtypes: STA (coverage_cell_mismatch, authority_unattested_binding); SPICE (corner/load/metric authority misbind).

Minimal-component ablation (exploratory; not headline). No stable minimal component underlies BundleS (C1+C2+C4+C7): C1 alone did not eliminate axis errors; C2-only and C4-only did not reproduce the bundle pattern; a predeclared in-window C24 bridge failed its ≥3/4-with-0-axis threshold (2/4, 1 axis) → `not_established`. The bundle-level effect is real within-family for Qwen3.7-Max; the C2×C4 joint-effect hypothesis remains unresolved.

B Study II — prospective STA per-instance detail

The per-instance table (Table 4) appears in the main text (Study II) as the authoritative STA evidence. The historical three-instance pilot (Base 0.50 / BundleS 0.33 / TypedContract 0.33) is descriptive and is **not pooled** with the prospective $n = 12$.

C Study III — Independent Benchmark Repair Audit (full 26-task coding)

Rubric frozen before coding; source evidence is PR #53 "Terminal-Bench 2.1" in the Terminal-Bench 2 repository (body sha256 `c7172fc6`; 57 files sha256 `fccf2d61`); 2.0/2.1 task snapshots frozen (89 tasks each; sha256 `5c7ed05c`). L1 cap=capability, L3 exec=execution, L4 art=artifact integrity.

Task	validity_threat	repair_mechanism	coverage
adaptive-rejection-sampler	L1 cap	instruction clarification	partial
build-pmars	L1 cap	instruction clarification; verifier/test tightening	partial
caffe-cifar-10	L3 exec; L1 cap	timeout; verifier/test broadening; instruction; env/dep; resource increase	partial
compile-compcert	L3 exec	environment/dependency repair	partial
configure-git-webserver	L1 cap	oracle repair; verifier/test tightening	partial
extract-moves-from-video	L3 exec	external-dependency removal	partial
filter-js-from-html	L1 cap; L3 exec	instruction clarification; resource increase	partial
fix-git	L4 art	hidden-artifact removal; verifier/test tightening	direct
hf-model-inference	L1 cap	verifier/test broadening	partial
install-windows-3.11	L1 cap	instruction clarification; verifier/test tightening	partial
make-doom-for-mips	L3 exec	oracle repair; environment/dependency repair	partial
mteb-leaderboard	L3 exec; L1 cap	env/dep repair; instruction; resource increase	partial
mteb-retrieve	L3 exec; L1 cap	env/dep repair; instruction clarification	partial
polyglot-c-py	L1 cap	verifier/test broadening	partial
polyglot-rust-c	L1 cap	verifier/test broadening	partial
protein-assembly	L1 cap	oracle repair	partial
rstan-to-pystan	L1 cap	oracle repair	partial
sam-cell-seg	L1 cap	instruction clarification	partial
torch-tensor-parallelism	L1 cap; L3 exec	instruction clarification; resource increase	partial
torch-pipeline-parallelism	L3 exec	resource increase	partial
query-optimize	L1 cap	instruction clarification	partial
train-fasttext	L3 exec; L1 cap	verifier/test tightening; env/dep repair	partial

build-pov-ray	Other	environment/dependency repair	not-covered
financial-document-processor	Other	environment/dependency repair	not-covered
mcmc-sampling-stan	Other	environment/dependency repair	not-covered
overfull-hbox	Other	environment/dependency repair	not-covered

Aggregate: direct 1 / partial 21 / not-covered 4; layers L1 17, L3 10, L4 1, Other 4; L2 sampling 0; 6 multi-layer.

D Operational independence and historical phase chronology

Five pre-registered structural criteria (independent text templates, role vocabularies, hidden-truth representations, grader implementations, decoy logic); operational, not a proof. Historical phase chronology (provenance; does not structure the main paper): discovery → transport repair → controlled pair → non-answer localization → frozen held-out → cross-model → cross-family (STA/SPICE) → TypedContract → prospective STA expansion. Episode counts 58/24/36/72; costs ¥682.25 / ¥7.72 / ¥11.75 / ¥43.56.

E Infrastructure, custody, and submission HEAD

Exact-commit isolated-worktree execution with pre/post-episode canonical-hash verification; committed sample-membership arbitration; SSE streaming; tool-health sentinel + full-path bookends; immutable-core/derived-deck action-surface repair; per-episode custody byte-matching; sanitized logs. A pre-flight caught and fixed a grading-path bug (malformed signoff-status JSON that would have forced a component false for every STA episode) before the prospective run; the historical pilot was unaffected (0/30 triggers). Exact submission HEAD and custody hashes are in the supplement.

Ethics statement. The preregistered blinded human construct-validity study (Study B) is *preregistered but unexecuted because qualified independent human annotators were unavailable*; no LLM annotator was substituted. We do not claim independent human validation of the semantic-binding taxonomy; construct validity rests on executable provenance/authority oracles.

AI-use statement (outside the page limit). An LLM-based coding/writing assistant was used for orchestration code, generators, infrastructure, report scripts, and prose drafting. All scientific results are produced by committed deterministic pipelines reproduced from frozen manifests; no claim was authored or validated by an LLM as an oracle; all numbers were generated by committed code and re-extracted from committed ledgers.

Reproducibility statement. Deterministic generators, independent graders, frozen manifests, schedules, custody hashes, and sanitized episode evidence are in the anonymized supplement. ICLR 2027 deadlines (per the official site): abstract September 18, 2026 AOE; full paper September 25, 2026 AOE.